HAUTE ÉCOLE D'INGÉNIERIE ET DE GESTION DU CANTON DE VAUD

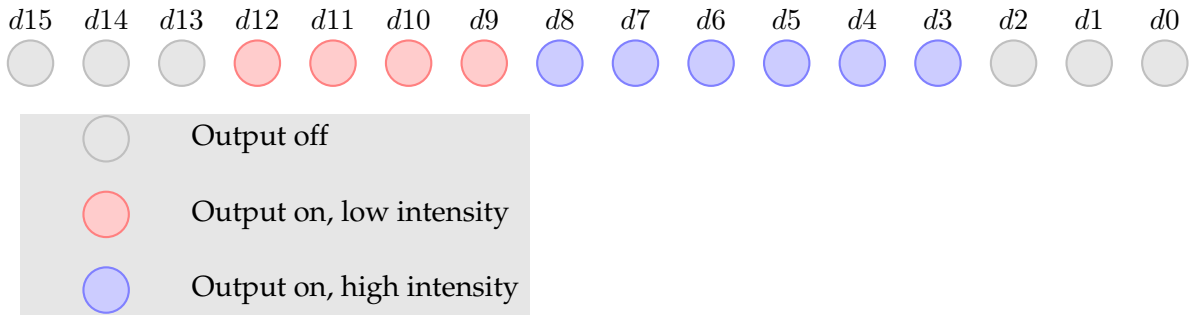
Prof. Yann Thoma

Formal verification
Verification of a linear display
CERN training - November 2026

Module to verify

We want to test a system that allows the linear display of a value. The display will be controlled to indicate where this value is in relation to two limits, Min and Max.
The example below shows the output for the following values :

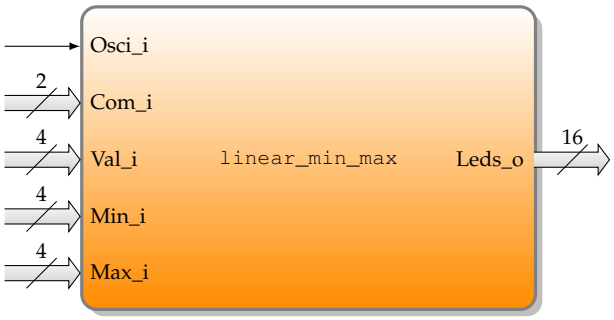
- `Max_i = 12`
- `Min_i = 3`
- `Val_i = 8`



Input/output

The input/outputs are :

Name	Size	Dir.	Description
`Com_i`	2	in	Mode of operation, on 2 bits
`Val_i`	4	in	Input value to be displayed on the linear display
`Min_i`	4	in	Lower bound
`Max_i`	4	in	Higher bound
`Osc_i`	1	in	Oscillatory signal for obtaining a low intensity
`Leds_o`	16	out	Linear display to be connected to 16 leds



Behavior

A LED at high intensity corresponds to constant activation of its corresponding output. A LED at low intensity is achieved by alternating 1-0 activation of the LED with the `Osci_i` signal.

Thus, the emitted power is lower. Therefore, in the case of low intensity, the activation is combined with the `Osci_i` signal thanks to an `and` gate.
The following table presents the mode of operation :

Command	Function	Description
00	Normal mode	If `Val_i` is in the interval $[Min_i, Max_i]$ — LEDs from `Min_i` to `Val_i` are on at high intensity — LEDs from `(Val_i+1)` to `Max_i` are on at low intensity — All other LEDs are off If `Val_i` is out of $[Min_i, Max_i]$, then all LEDs are off
01	Linear mode	Display of `Val_i` as a linear value, LEDs from `0` to `Val_i` are on with high intensity.
10	All off mode	All LEDs are off (`'0'`)
11	All on mode	All LEDs are on at high intensity (`'1'`)

⚠ When in normal mode, Max shall be greater than Min. If this is not the case, the output is undefined !
A generic parameter `ERRNO` allows to inject errors in the design. Its behavior is the following :

1. When in the $[0, 15]$ interval the result is valid ;
2. When in the $[16, 21]$ interval, the result is invalid.

This generic parameter allows to test your assertions by trying various `ERRNO` values thanks to modifications to the `.sby` file.